



Fieldnote

Entries made in the moment, even offline.
No confirmed entry is lost.

THE PROBLEM, IN PAPER DIARIES
11% actual versus 90% reported

Participant

Android app

- Joins by QR, code or invitation
- Text, photo, video, audio, scale
- Records offline, no duplicates
- Optional account, deletable data

Researcher

web portal

- Creates studies, activities and rules
- Invites: code, poster or pool
- Adherence and entries in real time
- Recorded time and received time
- Tags and exports the data

Administrator

web portal

- Researcher accounts with TOTP
- All studies and the pool
- Tamper-proof audit log
- Deletion requests

Security and privacy

- pseudonyms (P-07)
- roles
- signed URLs
- EXIF removed
- audit
- delete on day 7



71% adherence

Offline: kept safe

Architecture

Tailscale · 2 Traefik gateways

API

PostgreSQL

RabbitMQ

Garage (S3)

Patroni, etcd and quorum queues. Each file has 3 copies. Keeps receiving if a node goes down.

Scheduler (AI)

Deciding when to ask for each activity is a constraint satisfaction problem (CSP).

- availability
- activity window
- minimum gap
- daily limit
- max. per 15 min

Backtracking, MRV, LCV, forward checking and AC-3. Approved by the AI lecturer.

Validated stack by PDS and SD

- App React Native · Expo · TypeScript · SQLite · Expo Push
- Web React · Vite · Python 3.12 · FastAPI · ffmpeg
- Data PostgreSQL 17 · Patroni · etcd · RabbitMQ · Garage
- Infra Compose · Tailscale · Traefik · Prometheus · Grafana

2026 calendar

- 9 Oct pitch
- 11 Oct 1st delivery
- 13 Nov prototype
- 18 Dec final delivery